QUIZ

How can we apply linear regression if we have qualitative (categorical) features (predictors)?

REVIEW OF THE PREVIOUS LECTURE

Interaction effects: Given features X_1, X_2, introduce $X_1 \times X_2$ as well.

Polynomial regression: Given feature X_1, introduce X_1^2 as new feature.

Qualitative features: One hot encoding.

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Qualitative features: One hot encoding.

Restricting model complexity through regularization:

LASSO: Minimize $RSS + \lambda \sum_{i=1}^{m} |w_i|$ Ridge: Minimize $RSS + \lambda \sum_{i=1}^{m} w_i^2$

Notes: λ determined via cross validation.

Scaling is important if we do regularization.

OTHER CONSIDERATIONS: OUTLIERS

Remember the goal is to minimize the quadratic loss function,

$$RSS = \sum_{i=1}^{n} (y_i - (w_0 + w_1 x_{i1} + w_2 x_{i2} + \cdots + w_m x_{im}))^2$$

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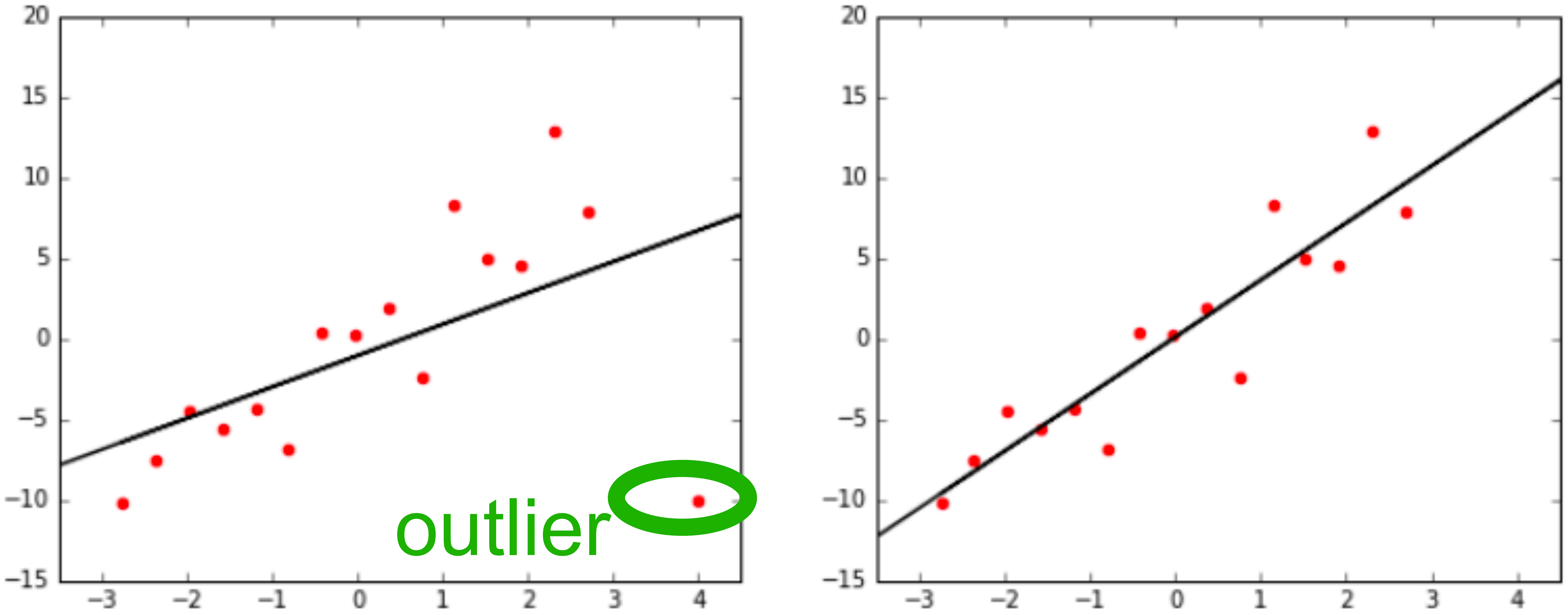
With quadratic error functions the effect of the outliers is strong:



Remove outliers, if you believe they reflect noise rather than signal.

SCALING THE DATA

Features vary over a wide range $\Rightarrow$ Coefficients vary over wide range

Ex: $$GDP = \$10{,}000 x_1 + 10{,}000{,}000{,}000{,}000 x_2$$

population

literacy rate

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Problems:

Interpretability: *Which feature is more important?*

Parameters of ML algorithms: *Constants that must hold for all variables*

(Step size in gradient descent with respect to which variable?)

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Problems:

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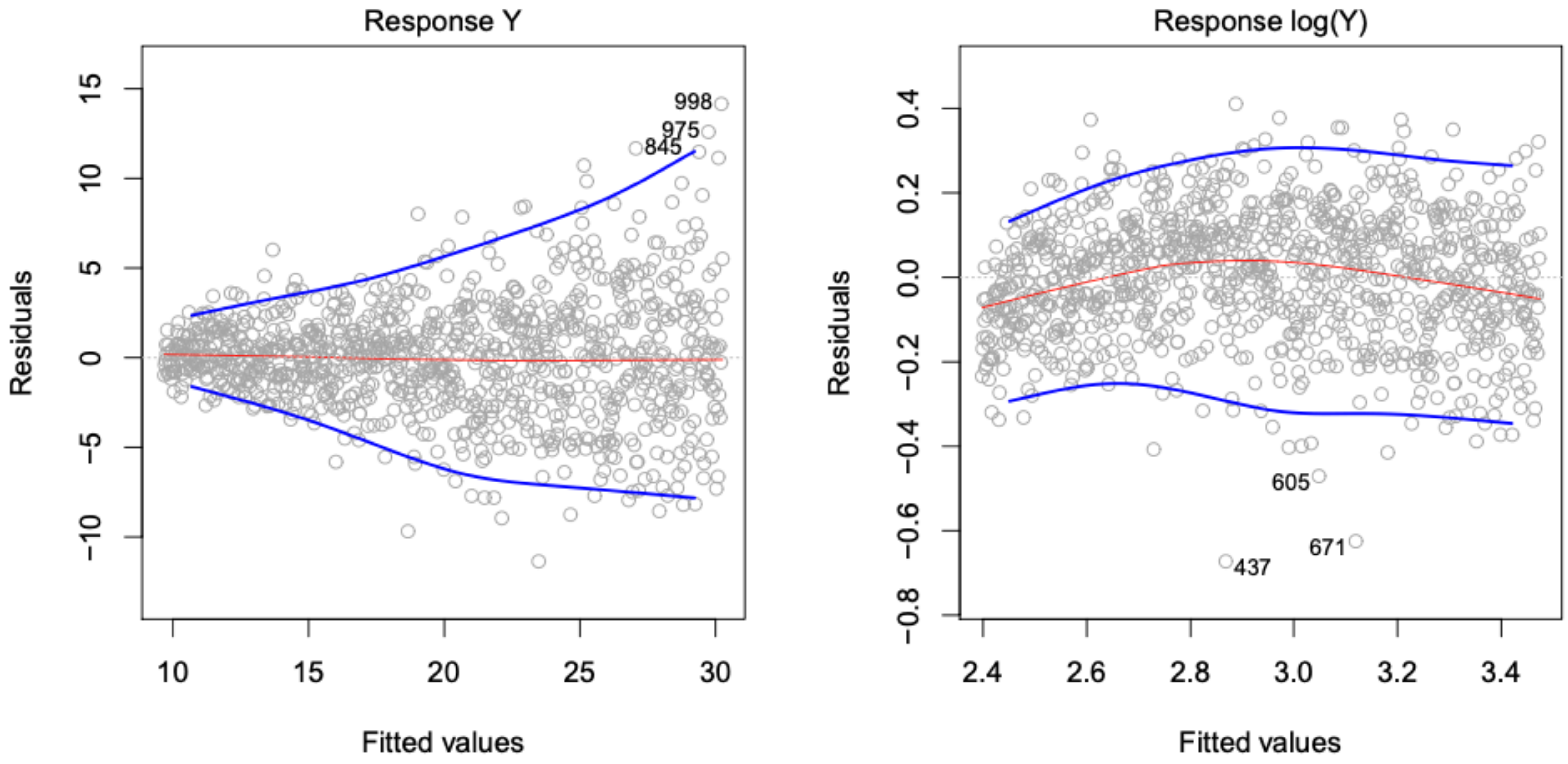
Solution: Bring them to the same scale by standardization ⇒ Z-score

(For feature j, $z_{ij} = \dfrac{x_{ij} - \overline{X}_j}{\widehat{\sigma_j}}$ where $\overline{X}_j$ is mean, $\widehat{\sigma_j}$ is standard deviation of feature j.)

HETEROSCEDASTICITY

SE, Confidence interval etc. rely on constant variance of error terms.

Sometimes variance of error terms not constant:



Transform response with $log Y$ or $\sqrt{Y}$.

DEALING WITH HIGHLY CORRELATED FEATURES

Having multiple features that are highly correlated is problematic.

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  How to interpret the coefficients?
- Hypothesis tests for $w_i = 0 \Rightarrow$ different results based on which specific linear combination used in the model.

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Solution:

- Check covariance matrix to find correlated features, eliminate one.
- OR combine into a single variable.

CLASSIFICATION WITH LOGISTIC REGRESSION

Training data: Number of hours studied for some course. We also have Pass (1) and Fail (0) response variable for the data points.

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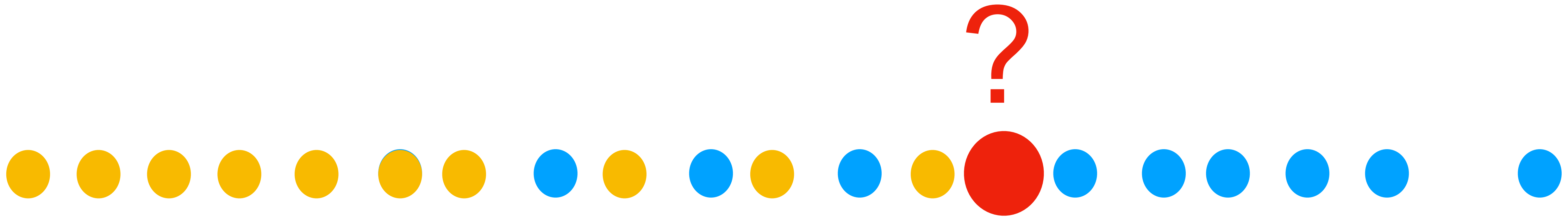
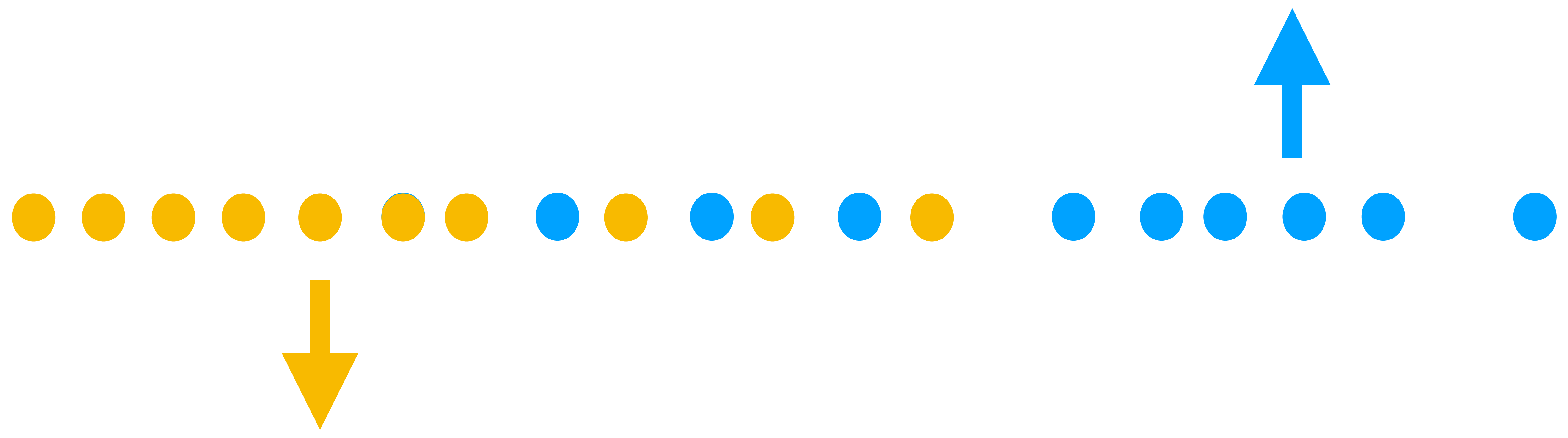


Can we train a model so that given a new data point (red) we can predict whether it corresponds to pass or fail?

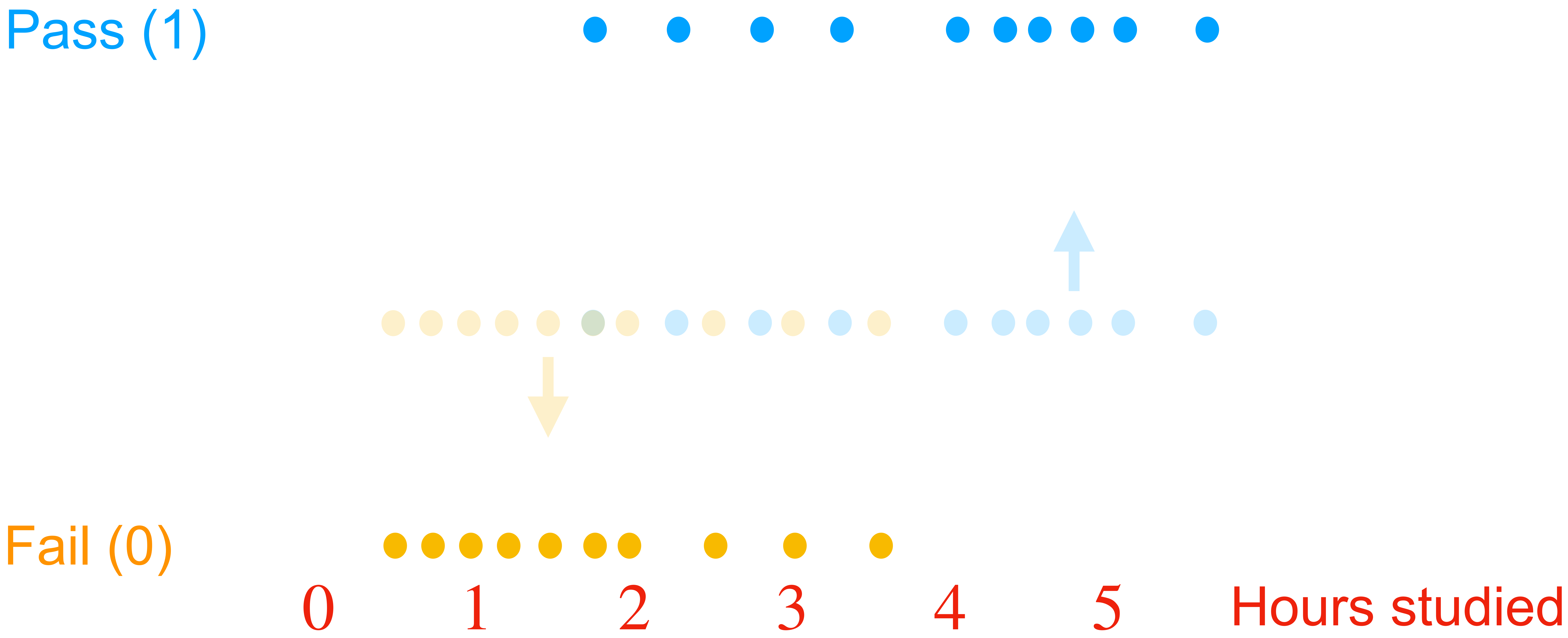
Different from value prediction: we now try to classify data point.

CLASSIFICATION WITH LOGISTIC REGRESSION



Let’s first separate them preserving X values.

CLASSIFICATION WITH LOGISTIC REGRESSION



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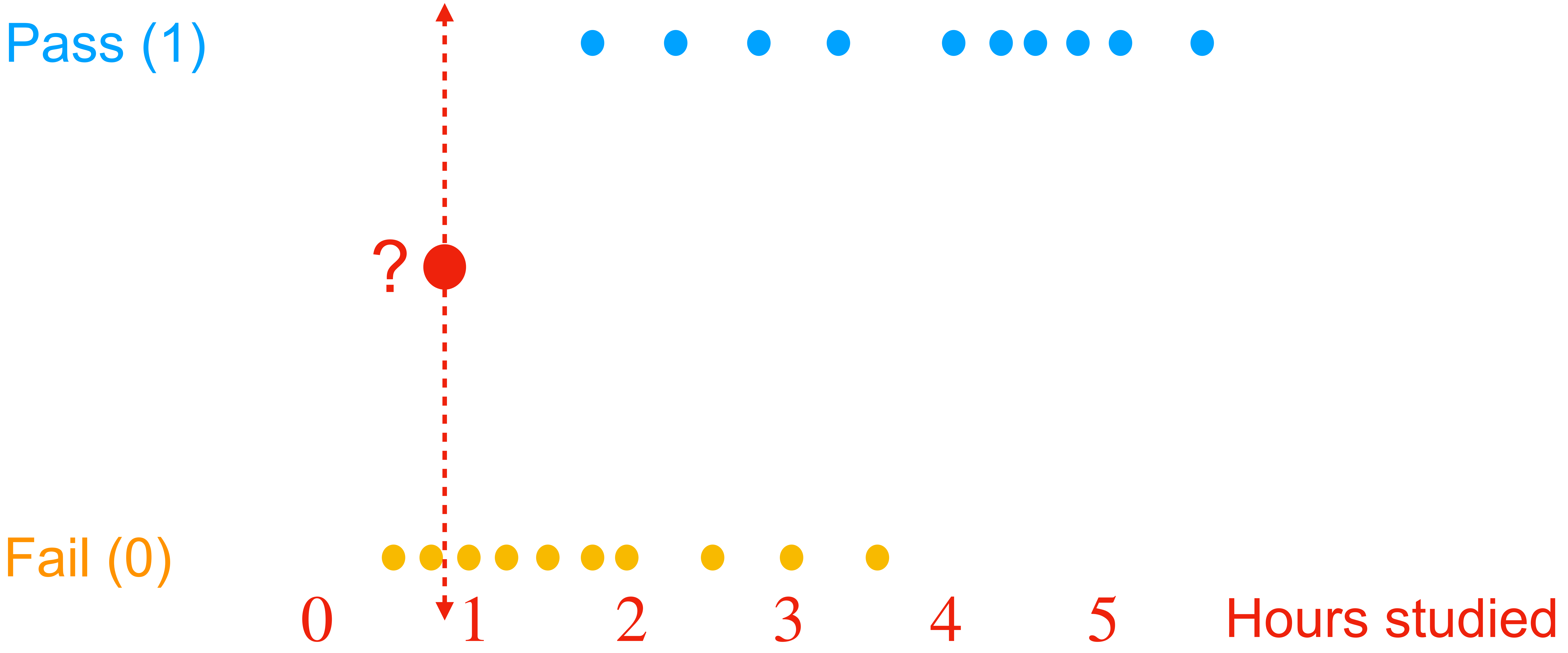

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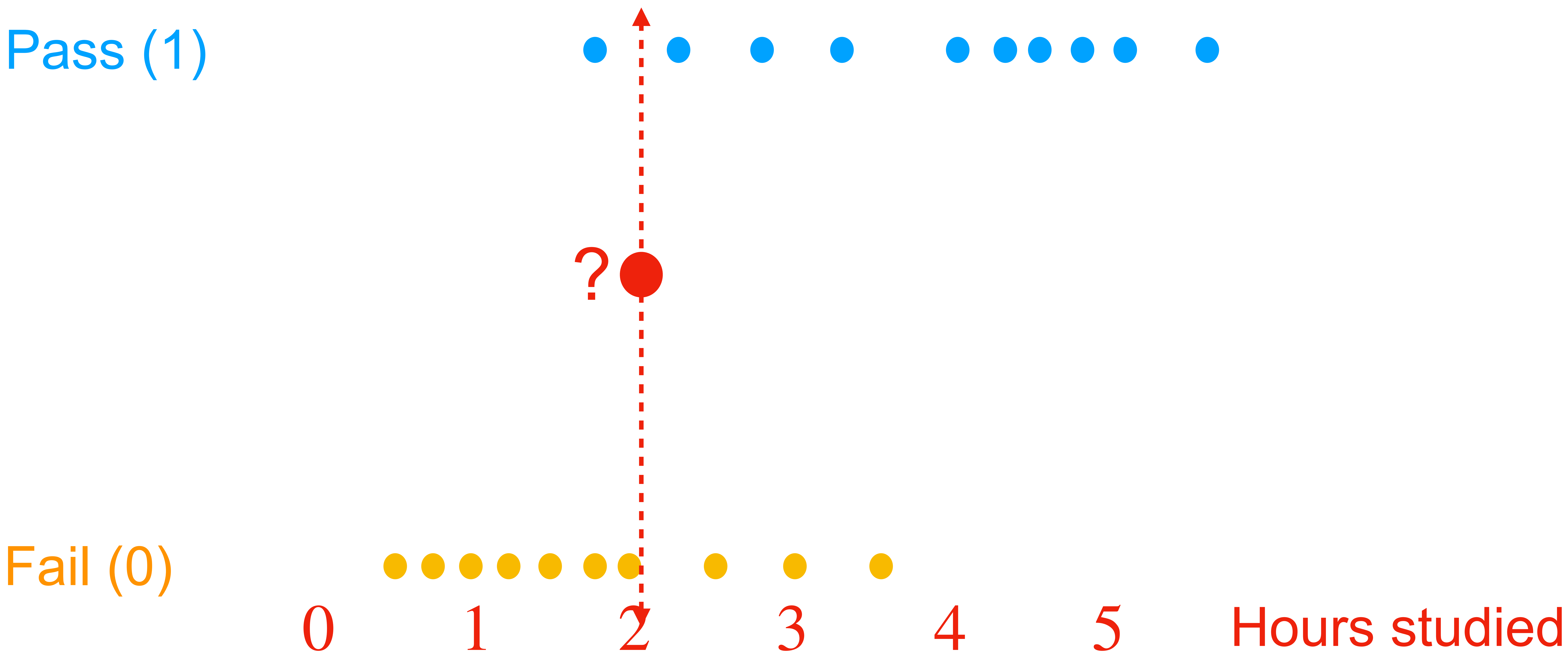

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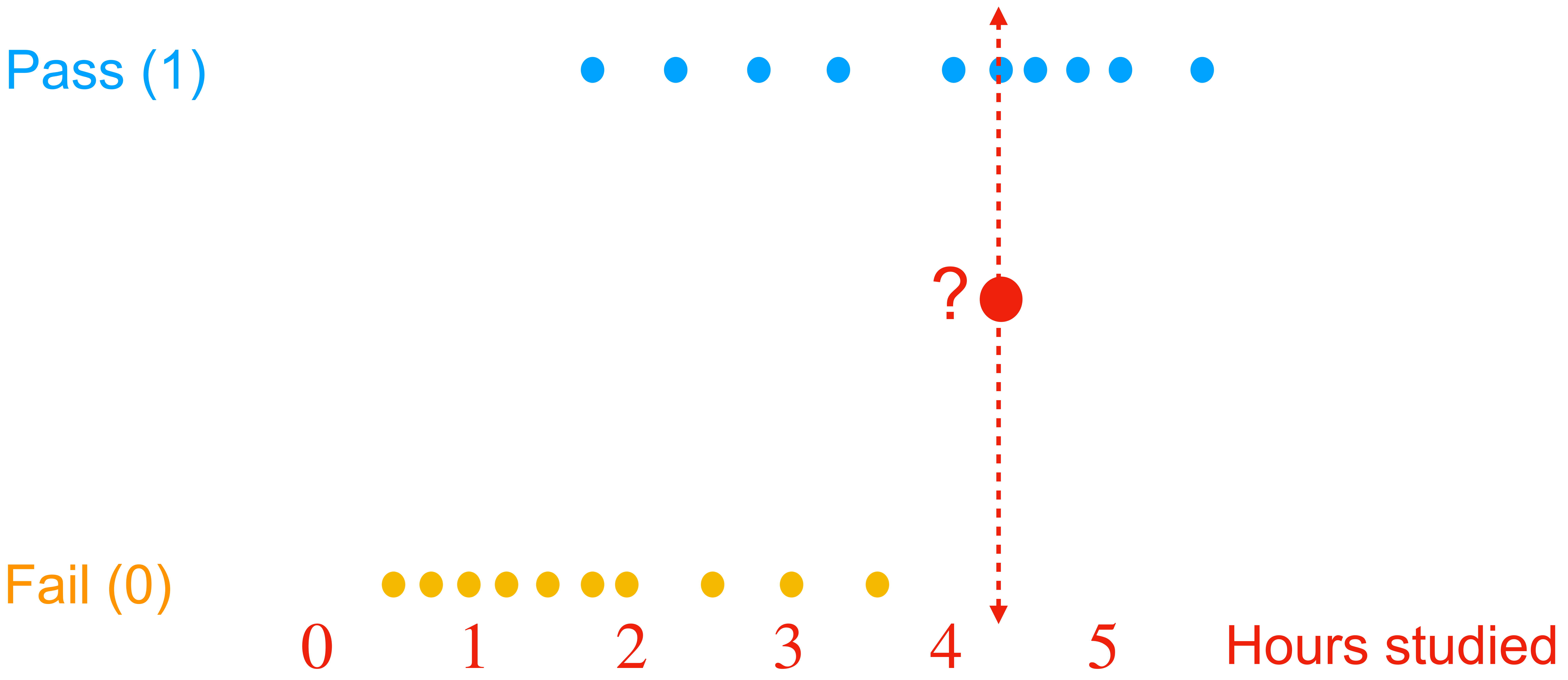
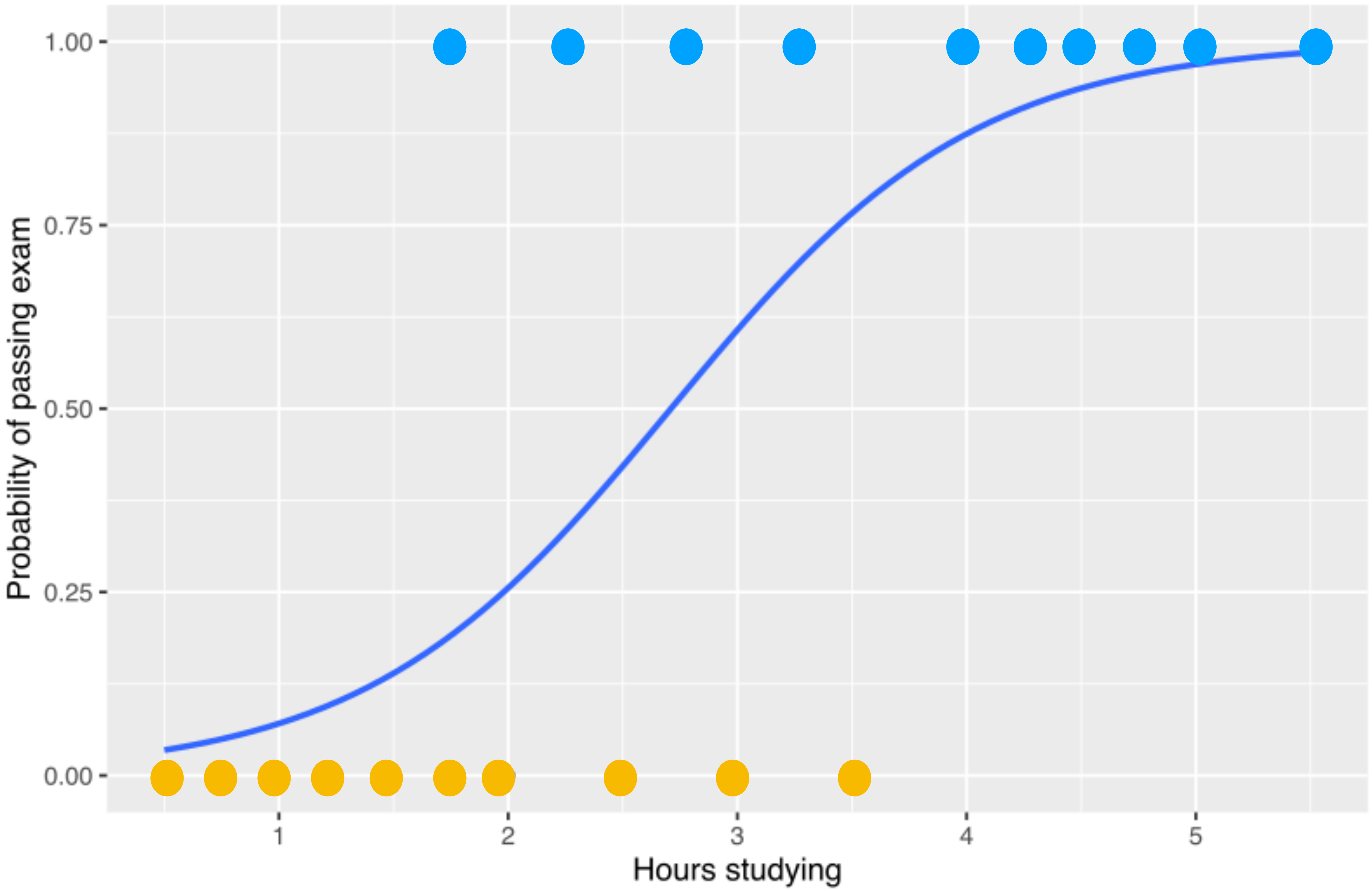


CLASSIFICATION WITH LOGISTIC REGRESSION

$Y = 1$

$Y = 0$



X: predictor (hours studying)

Y: response (pass or fail)

Blue curve plots:

$P(Y = 1 \mid X)$

CLASSIFICATION WITH LOGISTIC REGRESSION

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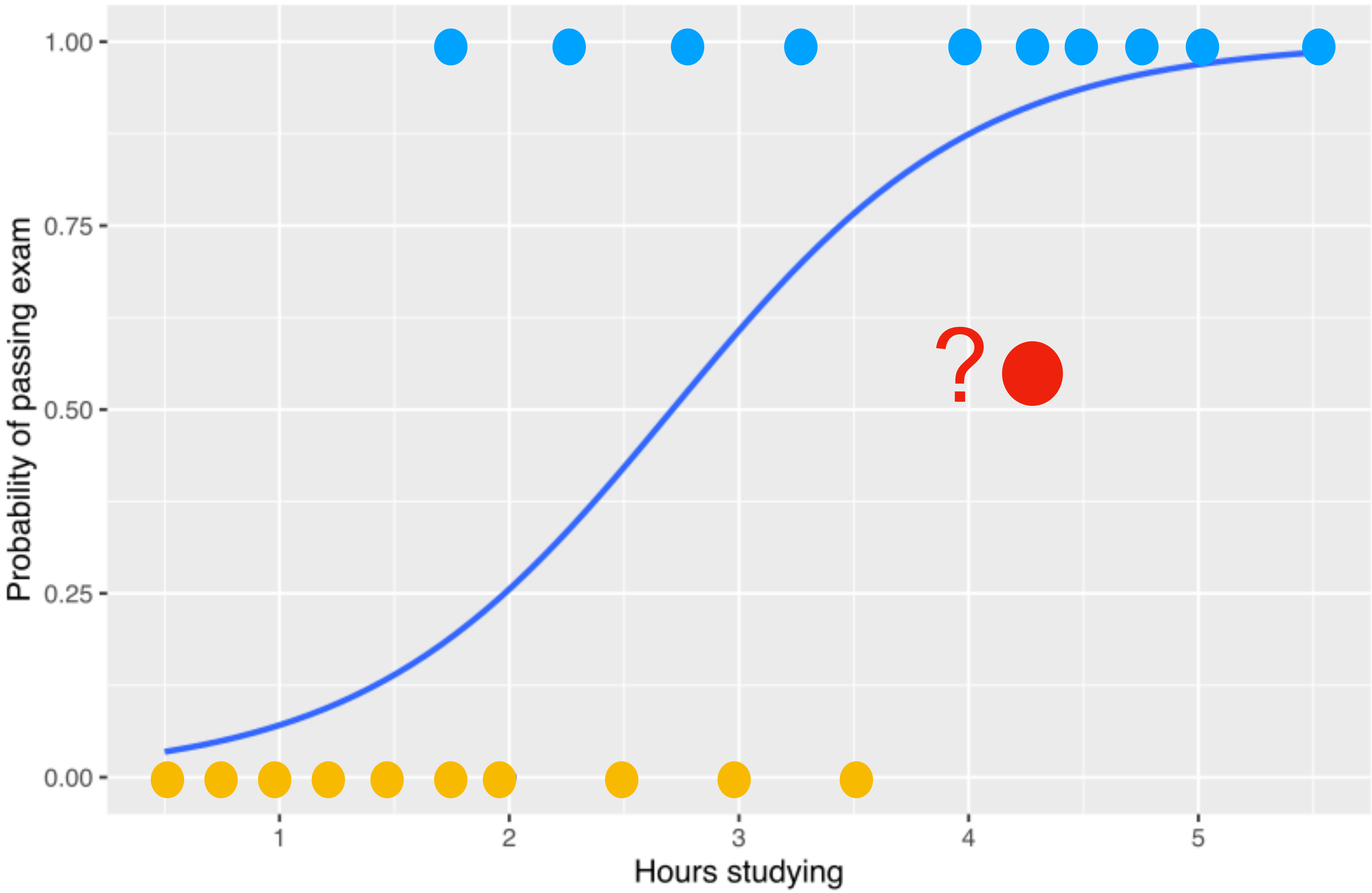

$Y = 0$

We can predict the class of red point based on blue curve:

if probability < 0.5
    predict fail (0)
else
    predict pass (1)

CLASSIFICATION WITH LOGISTIC REGRESSION

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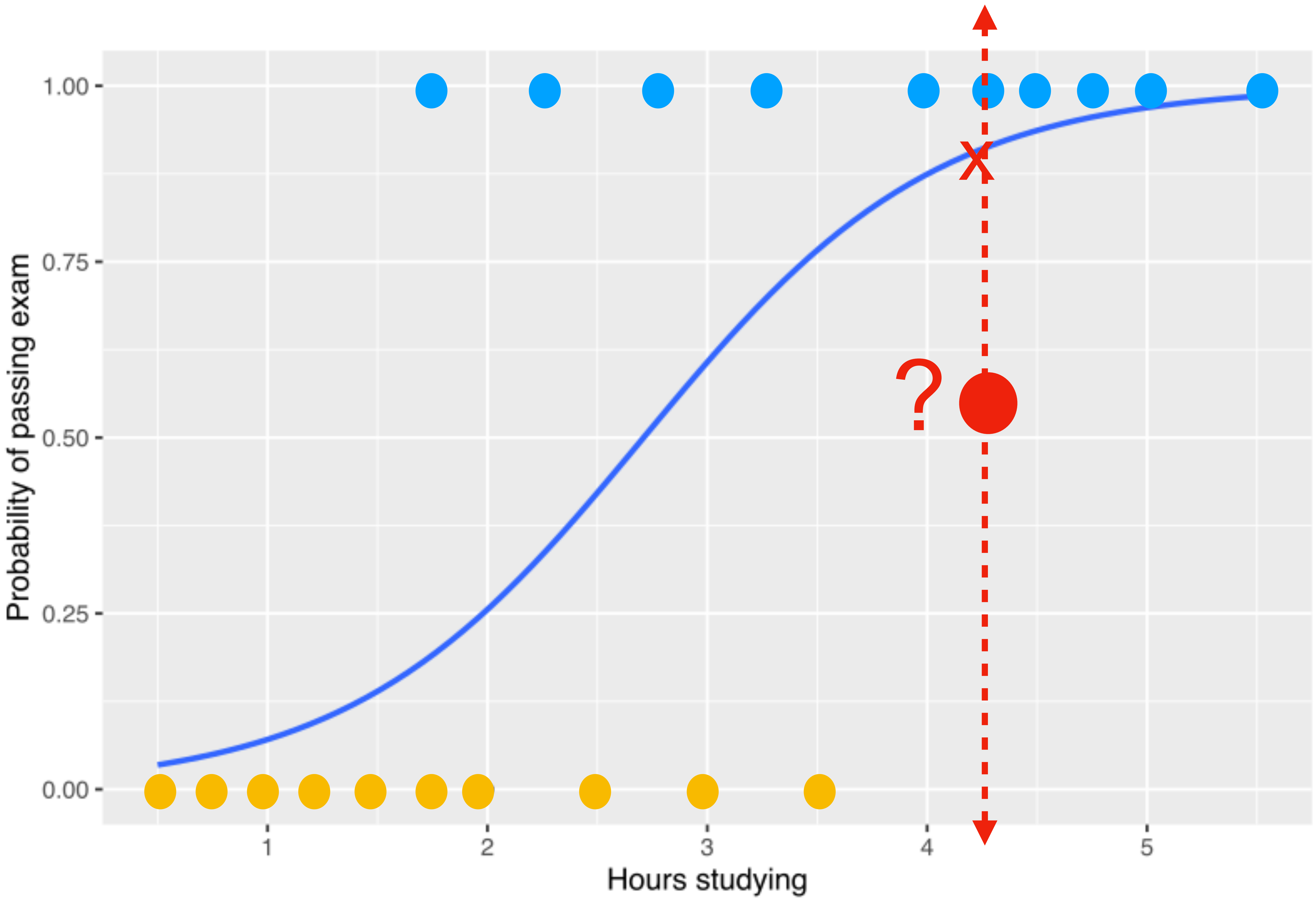

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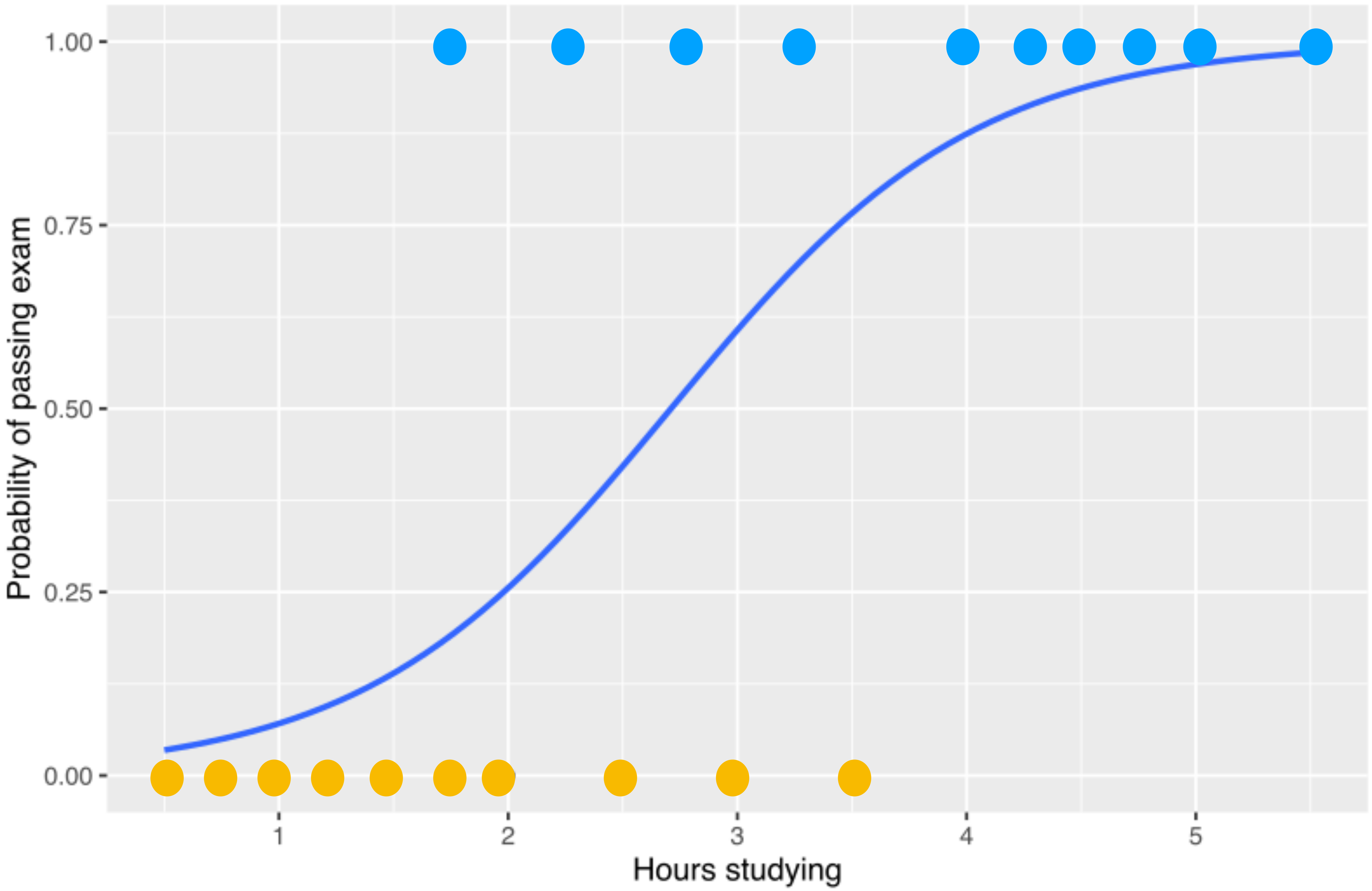
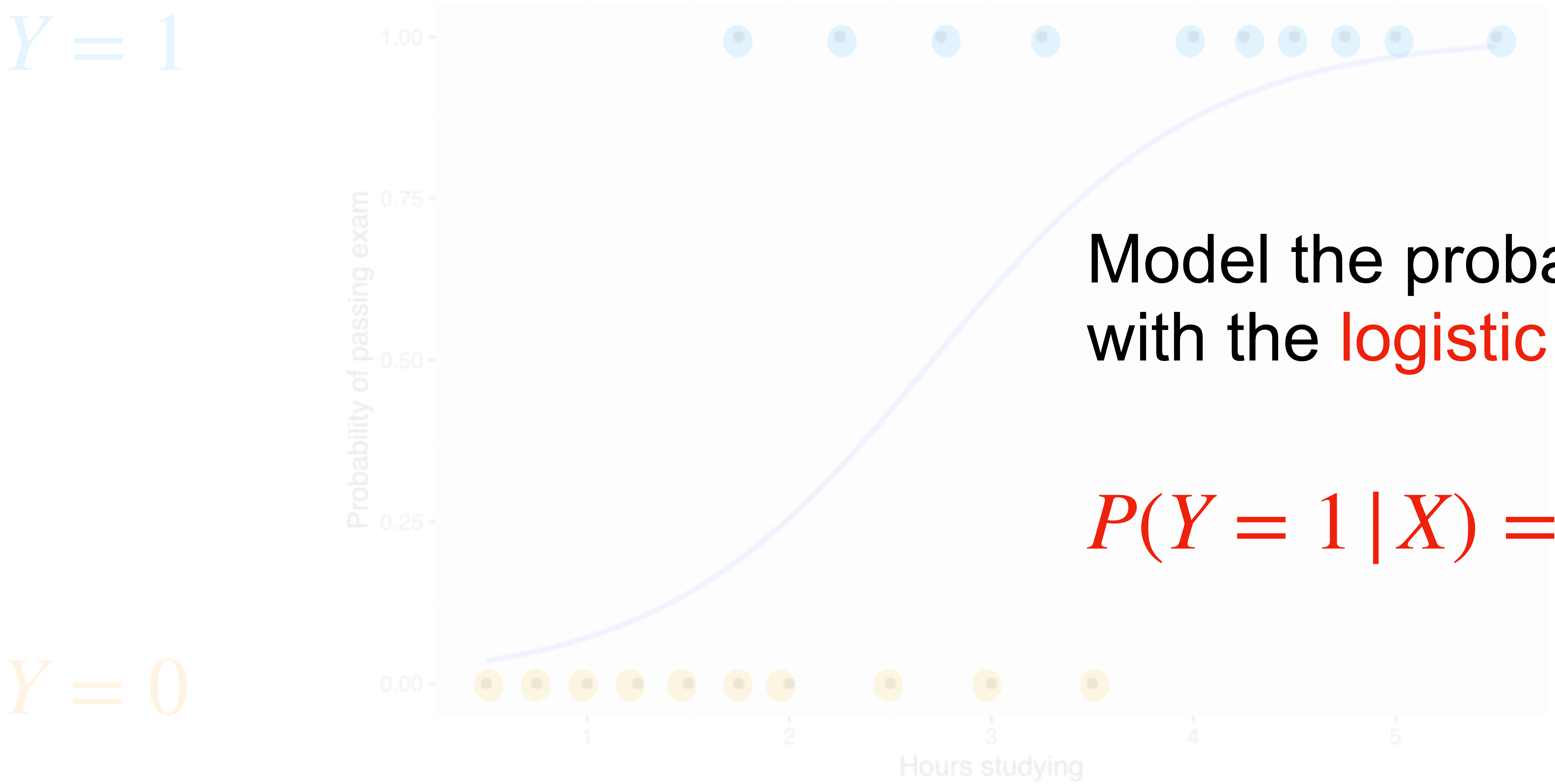


How to model the blue probability curve?

CLASSIFICATION WITH LOGISTIC REGRESSION



Model the probability curve with the logistic function:

$$P(Y = 1 \mid X) = \frac{1}{1 + e^{-(w_0 + w_1 x)}}$$

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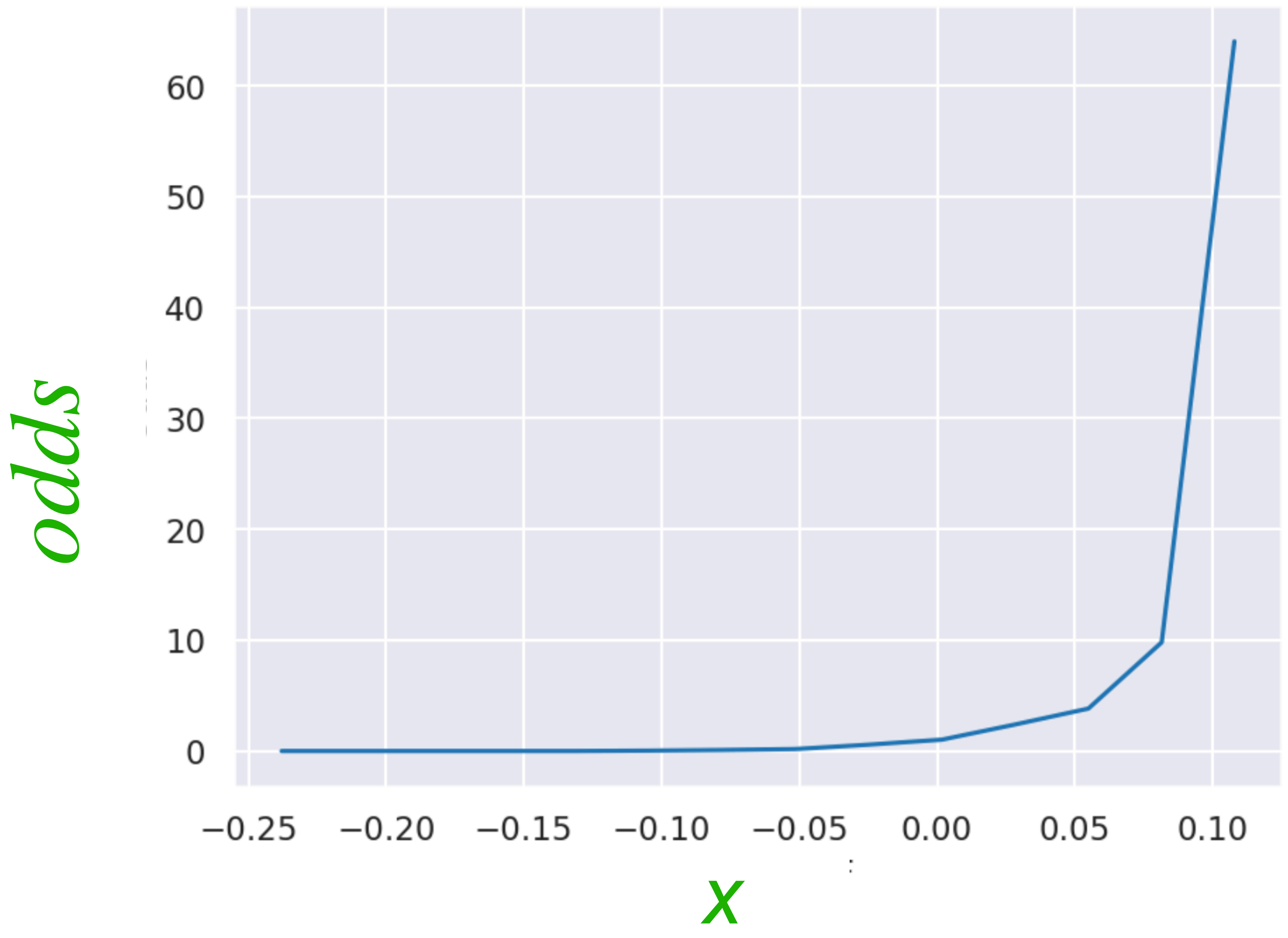

Intuitively, $0 \leq odds < 1?\ odds = 1?\ odds > 1?$

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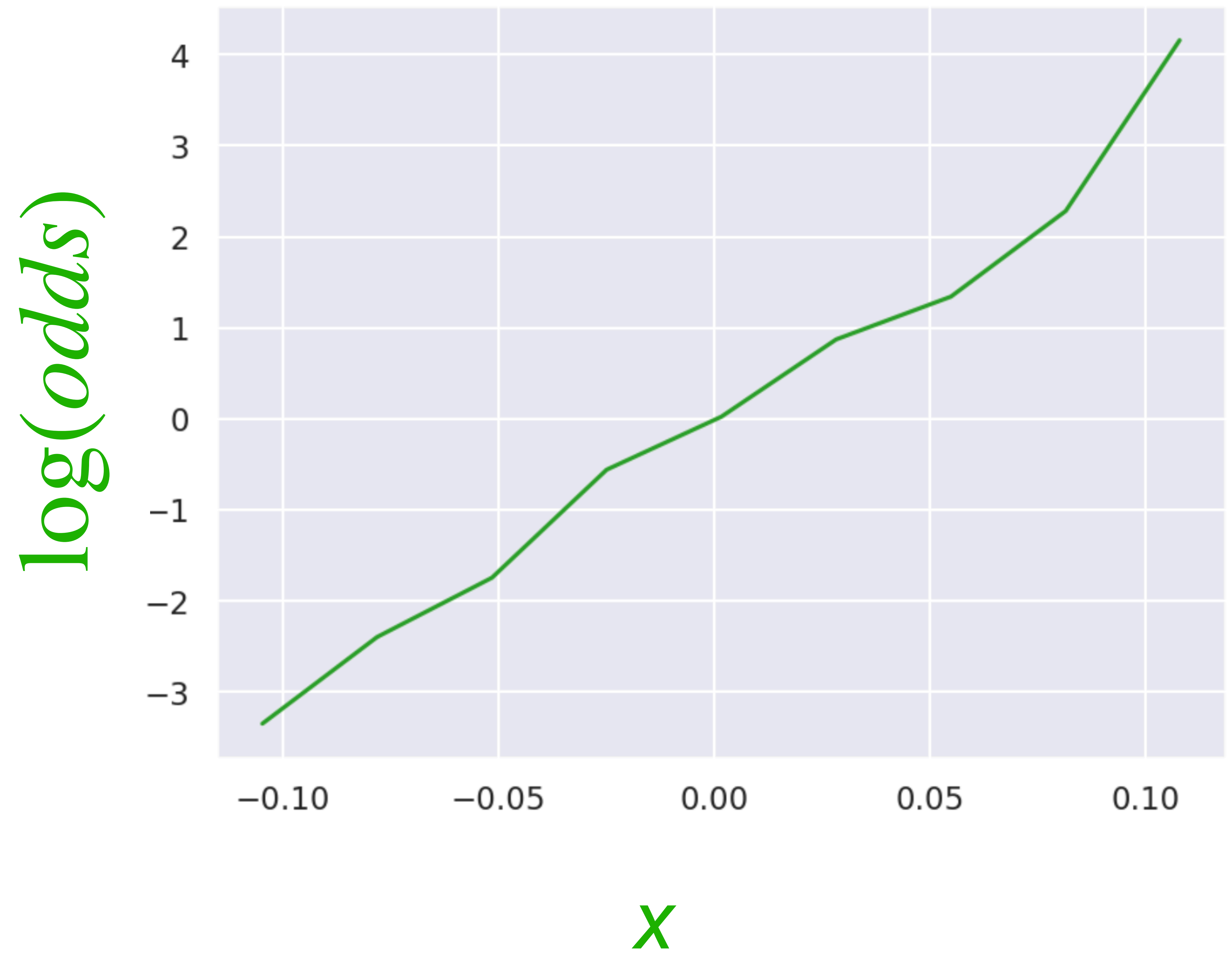
$$\log(odds) = \log(\frac{p}{1-p})$$



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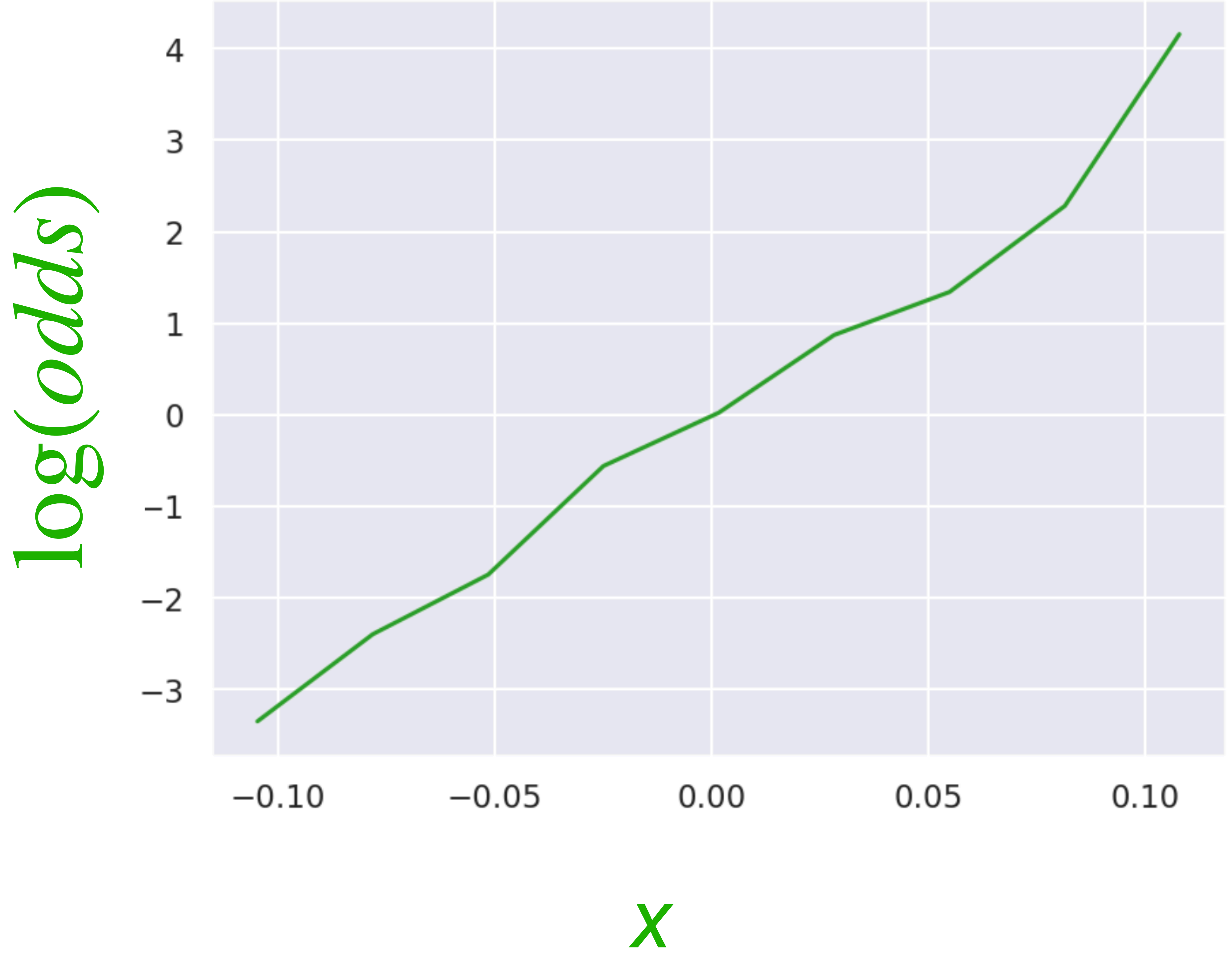

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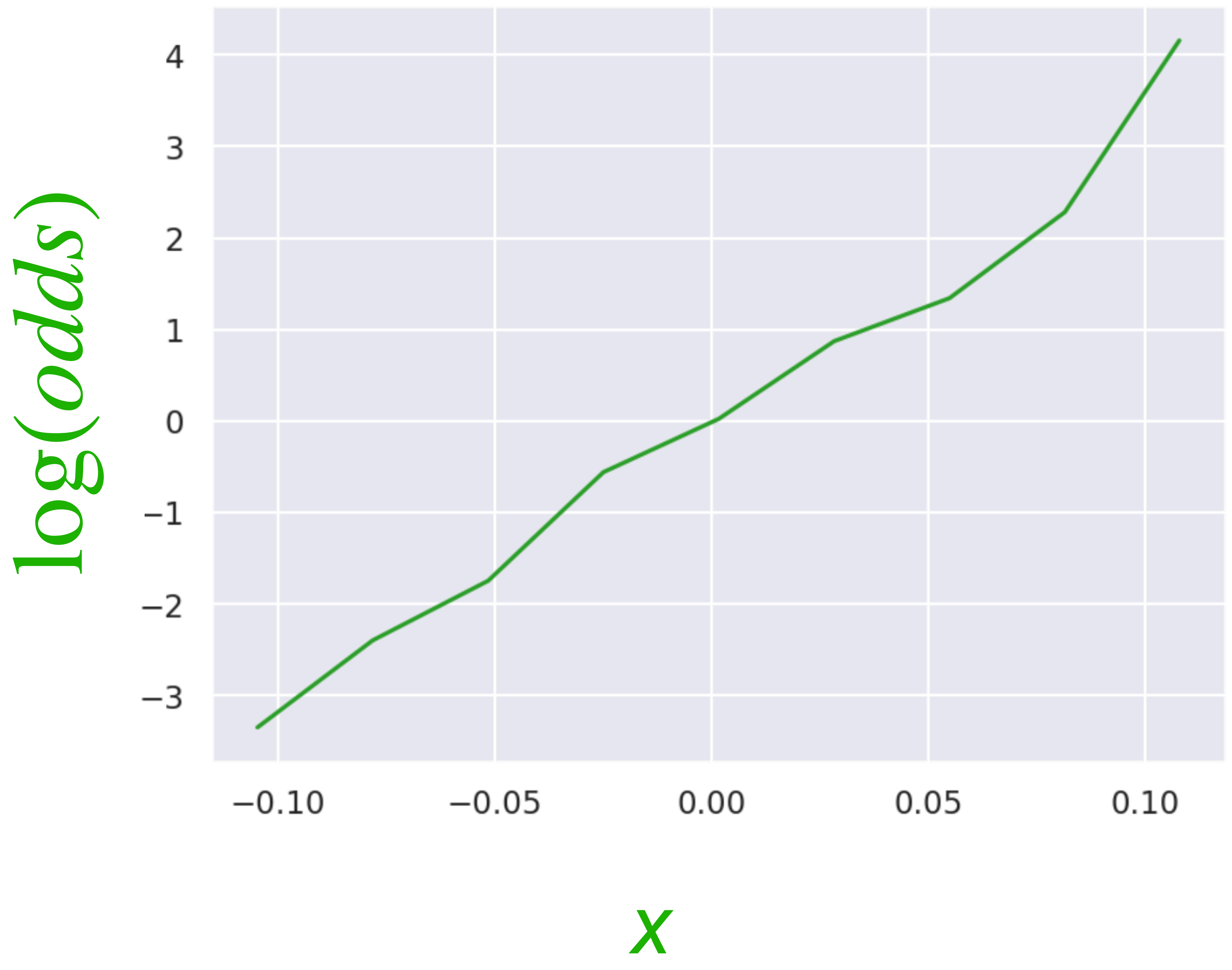

$$\log(\frac{p}{1-p}) = w_0 + w_1 x_1 \quad \Rightarrow \quad \frac{p}{1-p} = e^{w_0 + w_1 x_1}$$

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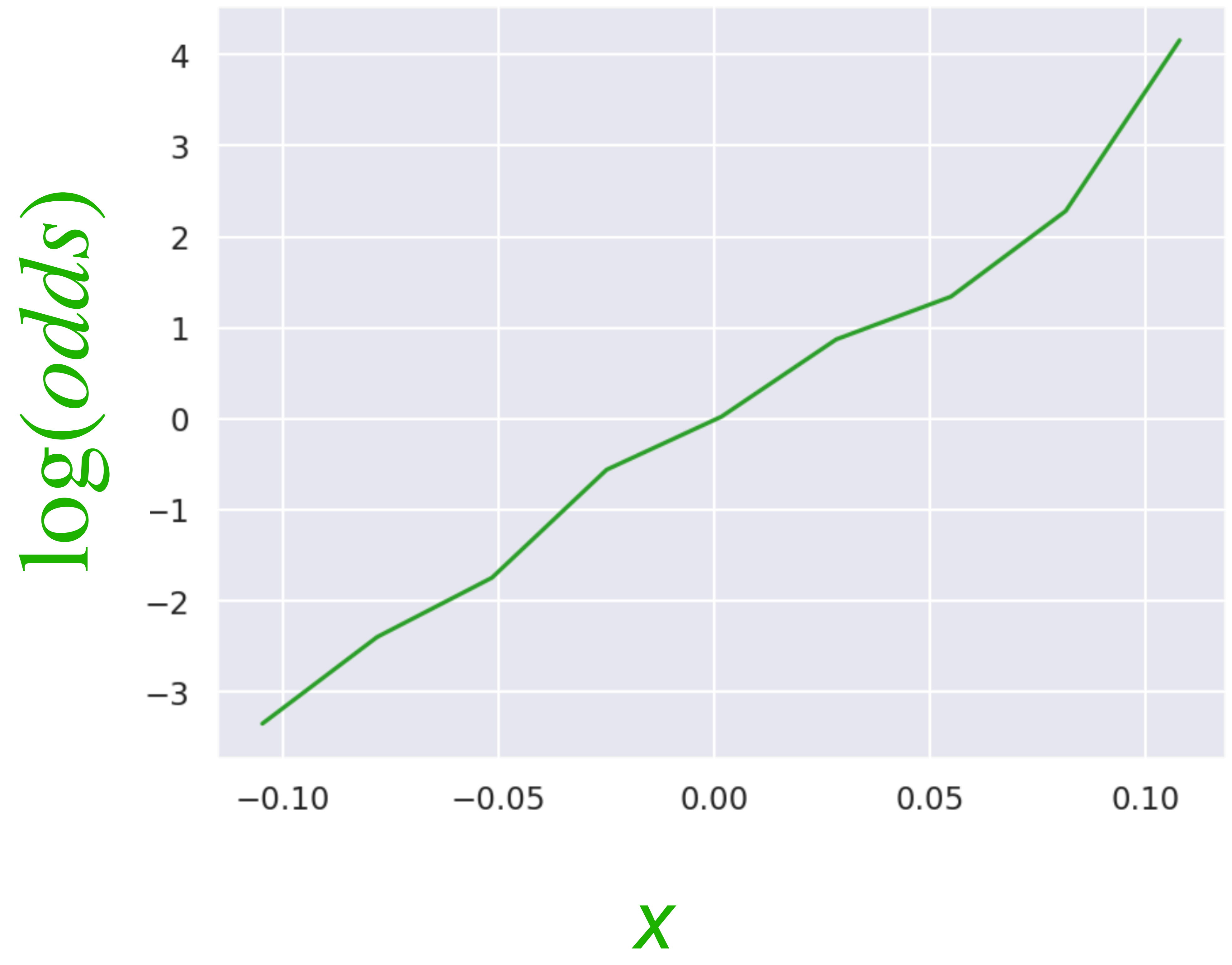

$$\log(\frac{p}{1-p}) = w_0 + w_1 x_1 \quad \Rightarrow \quad \frac{p}{1-p} = e^{w_0 + w_1 x_1} \quad \Rightarrow \quad p = \frac{1}{1 + e^{-(w_0 + w_1 x_1)}}$$

CLASSIFICATION WITH LOGISTIC REGRESSION

Ex: Say we fit the model and got $w_0 = -0.15, w_1 = 0.575$.

What is the probability that a new student who has studied for $x_1 = 2$ hours passes?

CLASSIFICATION WITH LOGISTIC REGRESSION

Ex: Say we fit the model and got $w_0 = -0.15, w_1 = 0.575$.

What is the probability that a new student who has studied for $x_1 = 2$ hours passes?

$$P(Y_1 = 1 \mid x_1) = \frac{1}{1 + e^{-(w_0 + w_1 x_1)}} = \frac{1}{1 + e^{-(-0.15 + 0.575 \times 2)}} = \frac{1}{1 + e^{-1}} \approx 0.73$$

CLASSIFICATION WITH LOGISTIC REGRESSION

Loss function for linear regression was

$$RSS = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2$$

where $\hat{y}_i = w_0 + w_1 x_i$

CLASSIFICATION WITH LOGISTIC REGRESSION

How about for logistic regression?

$$\boxed{\text{Loss function to minimize:} \quad -\sum_{i=1}^{n}\left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)}$$

Cross-entropy loss (CE)

where

$$p_i = P(y_i = 1) = \frac{1}{1+e^{-(w_0+w_1x)}}$$

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Cross-entropy loss (CE)

Intuitively, what is the error when:

$y_i = 1$ and $P(y_i = 1) \approx 1$?

$y_i = 1$ and $P(y_i = 1) \approx 0$?

$y_i = 0$ and $P(y_i = 1) \approx 1$?

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CLASSIFICATION WITH LOGISTIC REGRESSION

Where does the loss function come from?

Reminder:

Likelihood: Probability of observed data viewed as a function of the parameters of a statistical model.

$$\mathscr{L}(\theta \,|\, x) = P_\theta(X = x)$$

Maximum Likelihood Estimation: Estimate parameters that makes observed data most probable.

CLASSIFICATION WITH LOGISTIC REGRESSION

Where does the loss function come from?

First some simplification:

$P(Y_i = 1) = p_i$ and $P(Y_i = 0) = 1 - p_i$

Combine them into a single equation:

$$P(Y_i = y_i) = p_i^{y_i}(1 - p_i)^{1-y_i}$$

CLASSIFICATION WITH LOGISTIC REGRESSION

Where does the loss function come from?

Maximize $\displaystyle\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{1-y_i}$ Likelihood of data

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Maximize

$$\log\left(\prod_{i=1}^{n} p_i^{y_i}(1-p_i)^{1-y_i}\right) = \sum_{i=1}^{n} \log\left(p_i^{y_i}(1-p_i)^{1-y_i}\right) = \sum_{i=1}^{n} y_i \log(p_i) + (1-y_i)\log(1-p_i)$$

Minimize $$-\sum_{i=1}^{n} \left(y_i \log(p_i) + (1-y_i)\log(1-p_i)\right)$$